Compositional system

geometry and dynamics for,

United States — EMBER electricity generation

2001-2025

Report title	Compositional system geometry and dynamics for, United States — EMBER electricity
Generated	2026-05-06
Run ID	v1.1.x-final
Dataset ID	ember_usa
T (records)	25
D (carriers)	9
Engine version	cnt 2.0.4
Schema version	2.1.0
Engine signature (sha256[:32])	97748868a0381cb7b995d2ae562e5c42…
Source CSV SHA-256	ecdb50b25cb5807b38d4767a0344eddf…
Closed data SHA-256	81879347e313f033e024b2bc26041b8b…
Content SHA-256 (canonical JSON)	e6025305b686c55311edb31b49cb6d5d…

Reading order: geometry first, dynamics second.

```
  §A  Data disclosure & processing notes
  §B  Geometry : structure on the simplex
              evolution of proportions → ternaries → variation
              → correlation → scree → biplot → top-N biplot
              → balance dendrogram → SBP table
  §C  Dynamics : evolution through compositional space
              trajectories (Hs, ‖clr‖, ILR axes) → System Course Plot
              → bearing rose → bearing heatmap → ω → metric trace
              → helmsman → ring class → locks → CBS
              → pairwise divergence ranking → ω↔Hs
  §D  Summary
```

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Plate 2 | §A Data Disclosure & Processing-Anomaly Review

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
INPUT DATA
  source file        : ember_USA_United_States_generation_TWh.csv
  source SHA-256     : ecdb50b25cb5807b38d4767a0344eddf8390b8be9b58cb430b0a92ce4b13c3bd
  closed data SHA-256: 81879347e313f033e024b2bc26041b8b5004161e55cea4ee41859bf5003f7e0d
  records (T)        : 25
  carriers (D)       : 9
  carriers           : Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Other Renewables, Solar, Wind

ORDERING
  is_temporal        : True
  ordering_method    : by-time

ZERO REPLACEMENT
  zeros replaced     : 0
  replacement δ      : 1e-15
  method             : multiplicative

UNITS LINEAGE  (schema 2.1.0)
  input_units                 : ratio
  higgins_scale_units         : neper
  units_scale_factor_to_neper : 1.0

ENGINE CONFIG ACTIVE OVERRIDES
  (none — defaults from cnt.py USER CONFIGURATION block)

ENGINE SOURCE FINGERPRINT
  stage2_locked.py      : c318ab82c15890b947c3fc15a6c66b55aa23f3cf94104cf9b9d5fbdb
  cnt.py                : 1e10cb62d6588ac9557c6a2e8436c2980f0c35b379f29c05c0377711
  combined sha256       : 97748868a0381cb7b995d2ae562e5c429988ee7fd556943b882c5fcb

DEGENERACY FLAGS
  (none)

LOCK EVENTS  (period-2 detector)
  total events       : 2
    lock                  : 2

EITT BENCH-TEST RESIDUALS
  H_mean_full           : 1.5014923024404339
  M_sweep               : [{'M': 2, 'n_blocks': 13, 'H_mean_decimated': 1.50791723
  gate_pct              : 5.0
  note                  : Empirical observation of trajectory smoothness under tem

SANITY CHECKS  (computed during this report)
  ‖clr‖ non-finite count : 0
  ω    non-finite count : 0
  Hs   non-finite count : 0
  Σ CLR per row range   : [-3.55e-15, +7.99e-15]   (should be ≈ 0)
```

If any flag above is non-empty, the analysis on subsequent pages still ran but the reader should weight those plates accordingly. Determinism is preserved either way.

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Plate 3 | Evolution of Proportions (raw + cumulative)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

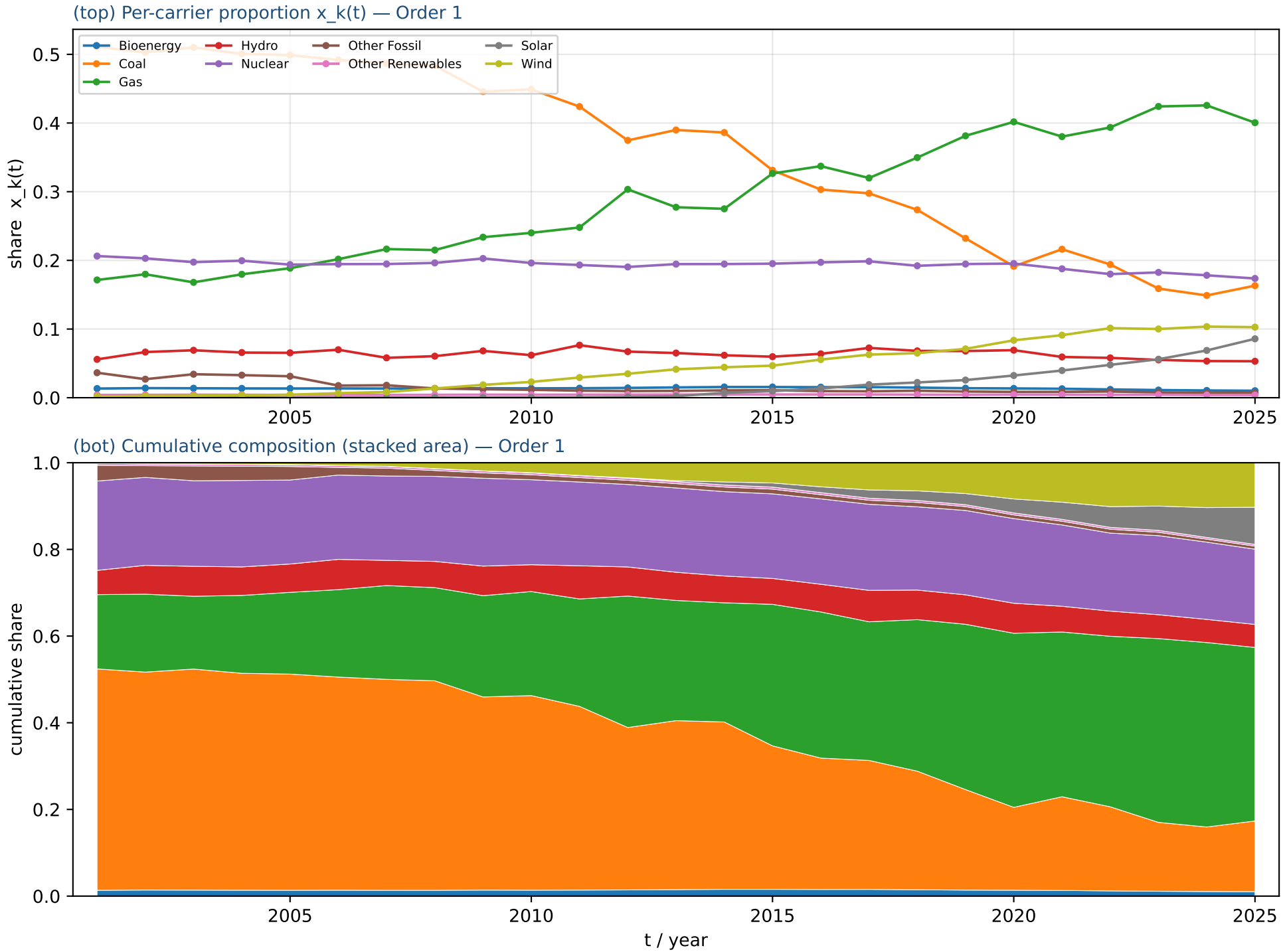

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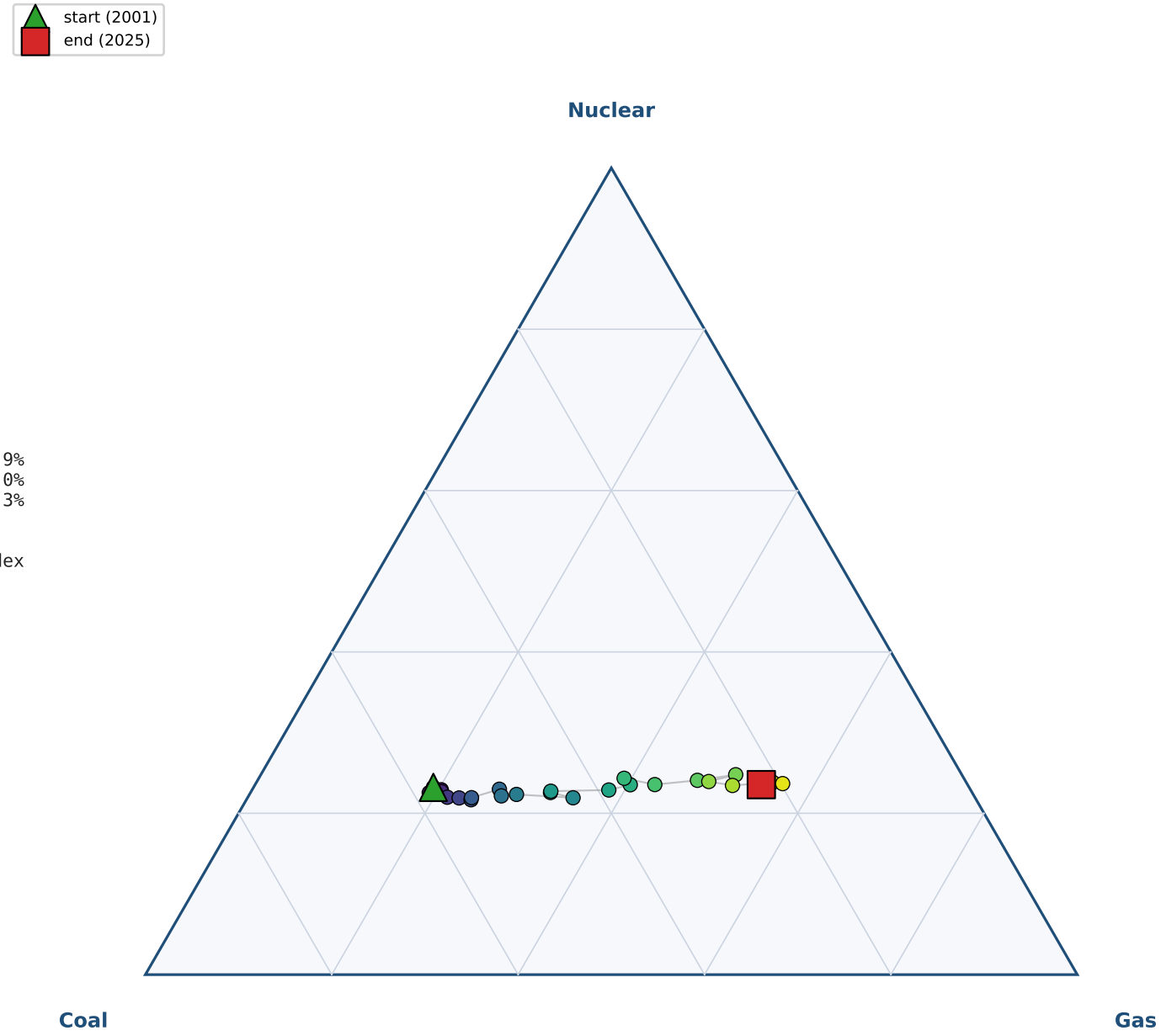
Plate 4 | Ternary Diagram (top-3 carriers)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

3-part subcomposition closed over Coal, Gas, Nuclear
(remaining 6 carriers absorbed into closure)




```
Top-3 by mean share:
  Coal            35.9%
  Gas             29.0%
  Nuclear         19.3%

T = 25 timesteps
colour map = time index
▲ start  ■ end
```

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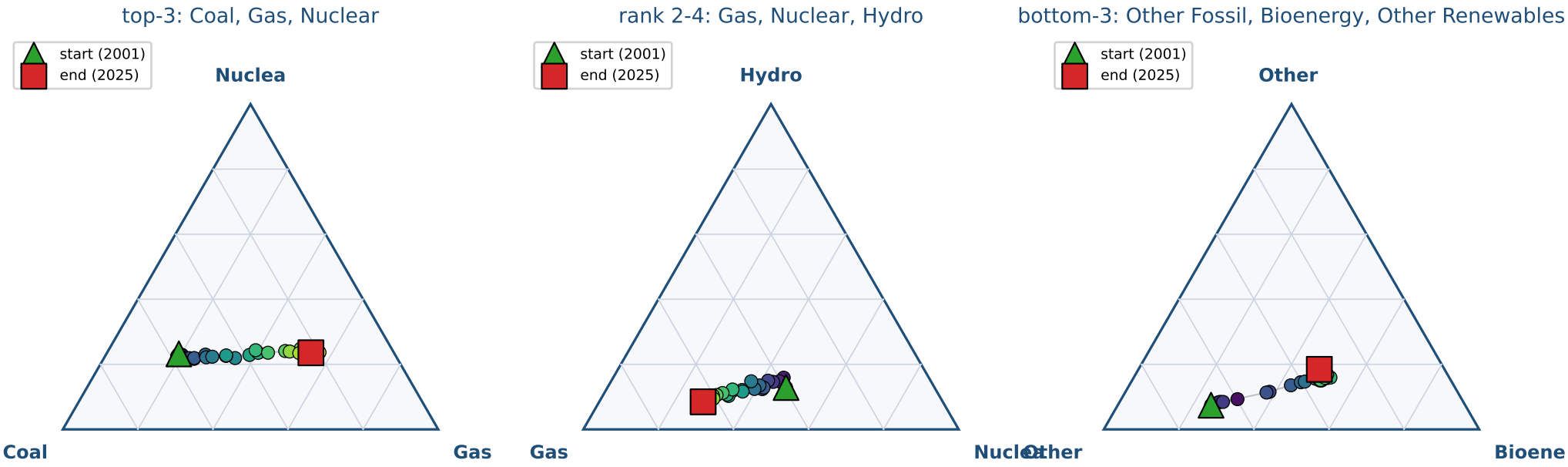
Plate 5 | Three-Part Subcompositions (3 ternaries)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: default heuristic

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Plate 6 | Variation matrix τ_ij heatmap

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

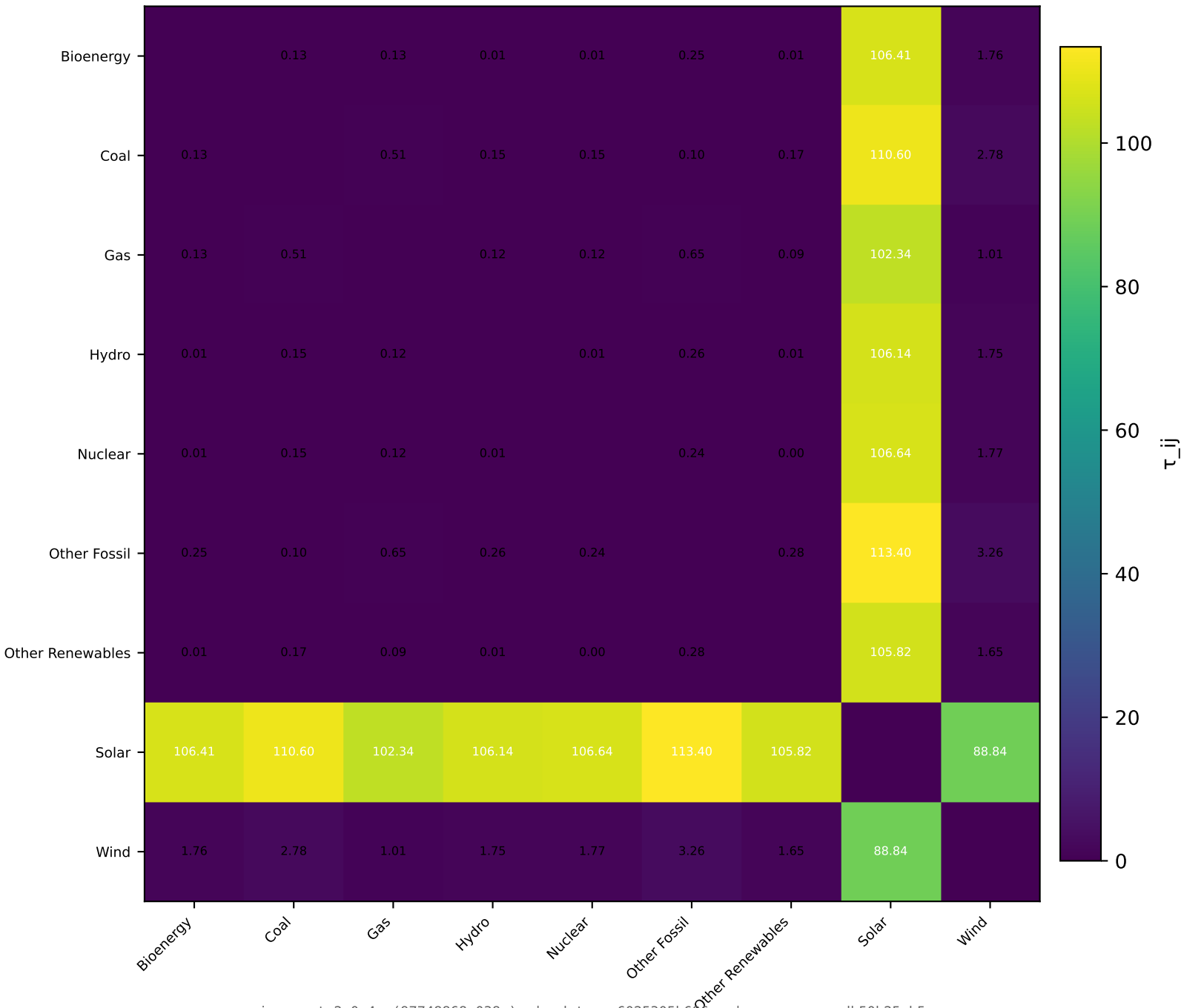

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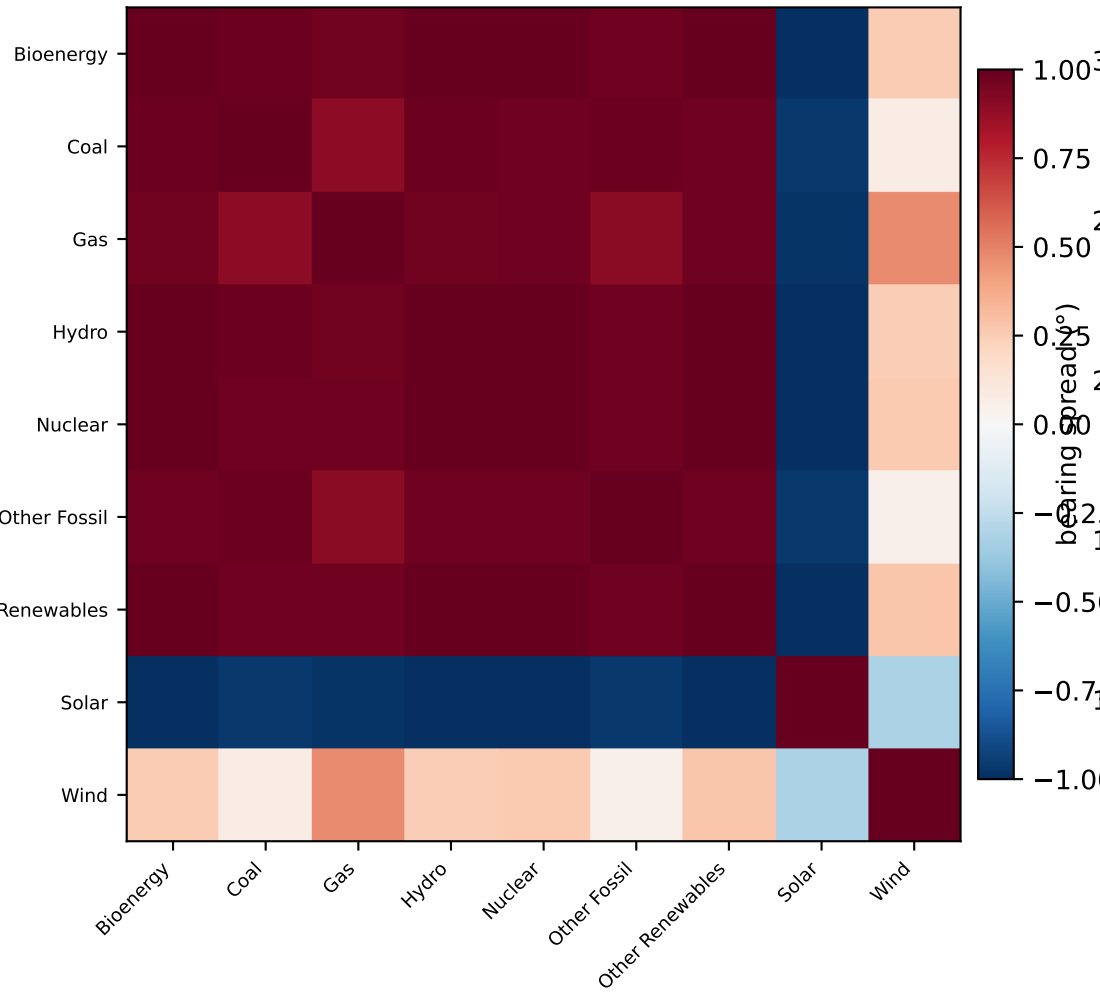
Plate 7 | Pair Pearson r heatmap + co-movement scatter

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



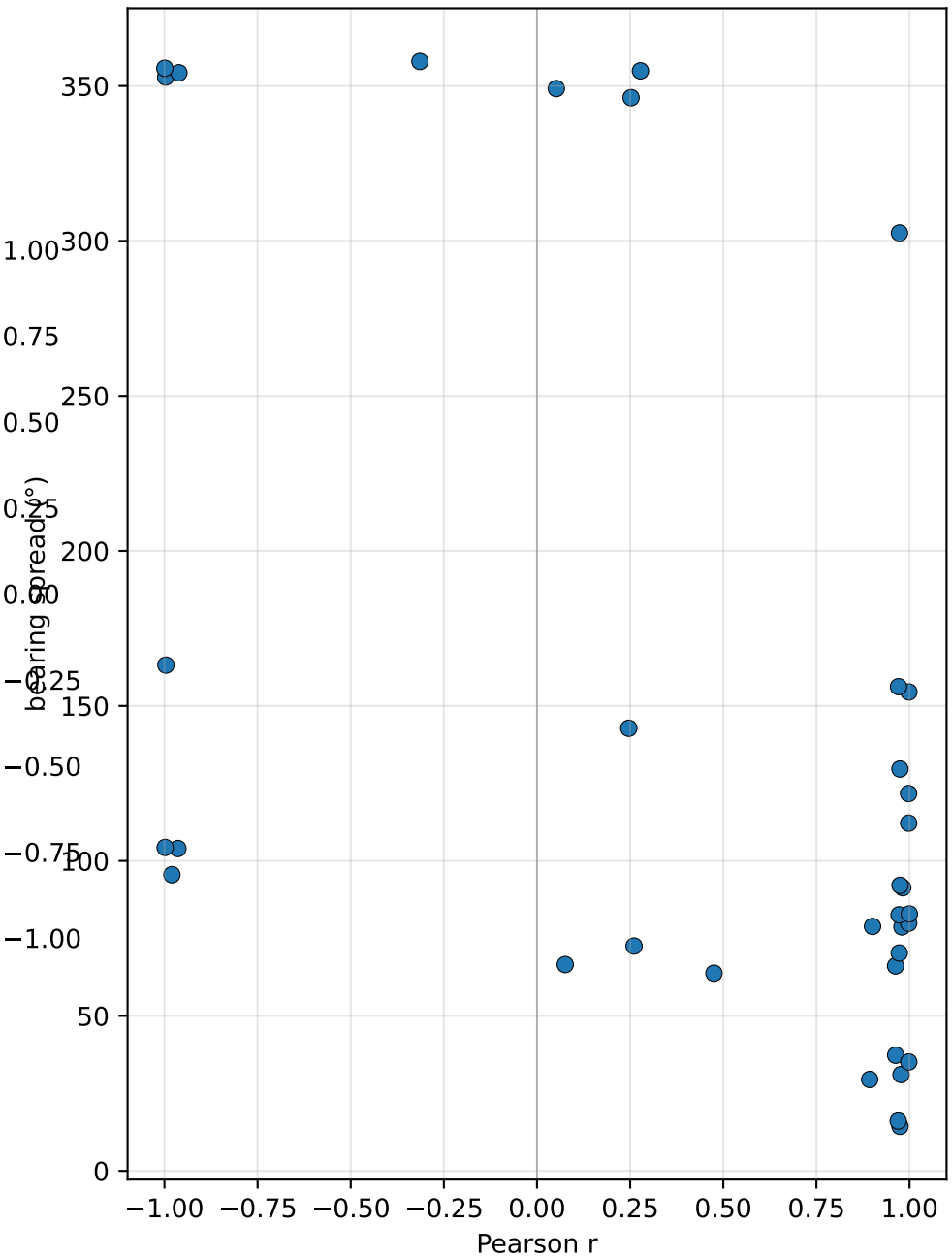


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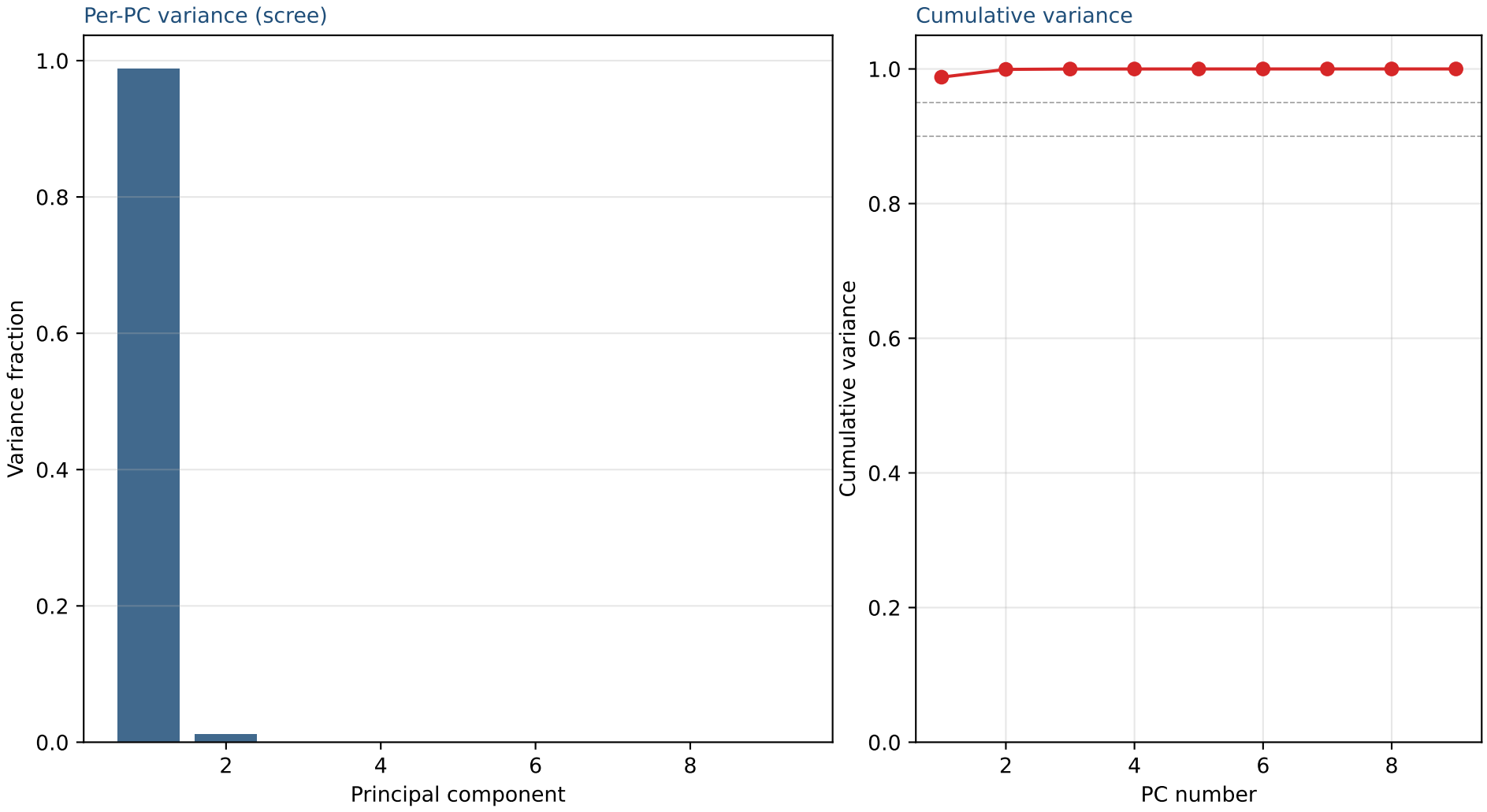
Plate 8 | Scree Plot - CLR PCA

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2




```
Per-PC variance contribution:
  PC1: 98.78%   cum 98.78%
  PC2:  1.15%   cum 99.93%
  PC3:  0.05%   cum 99.99%
  PC4:  0.01%   cum 99.99%
  PC5:  0.00%   cum 100.00%
  PC6:  0.00%   cum 100.00%
  ... 3 more PCs
```

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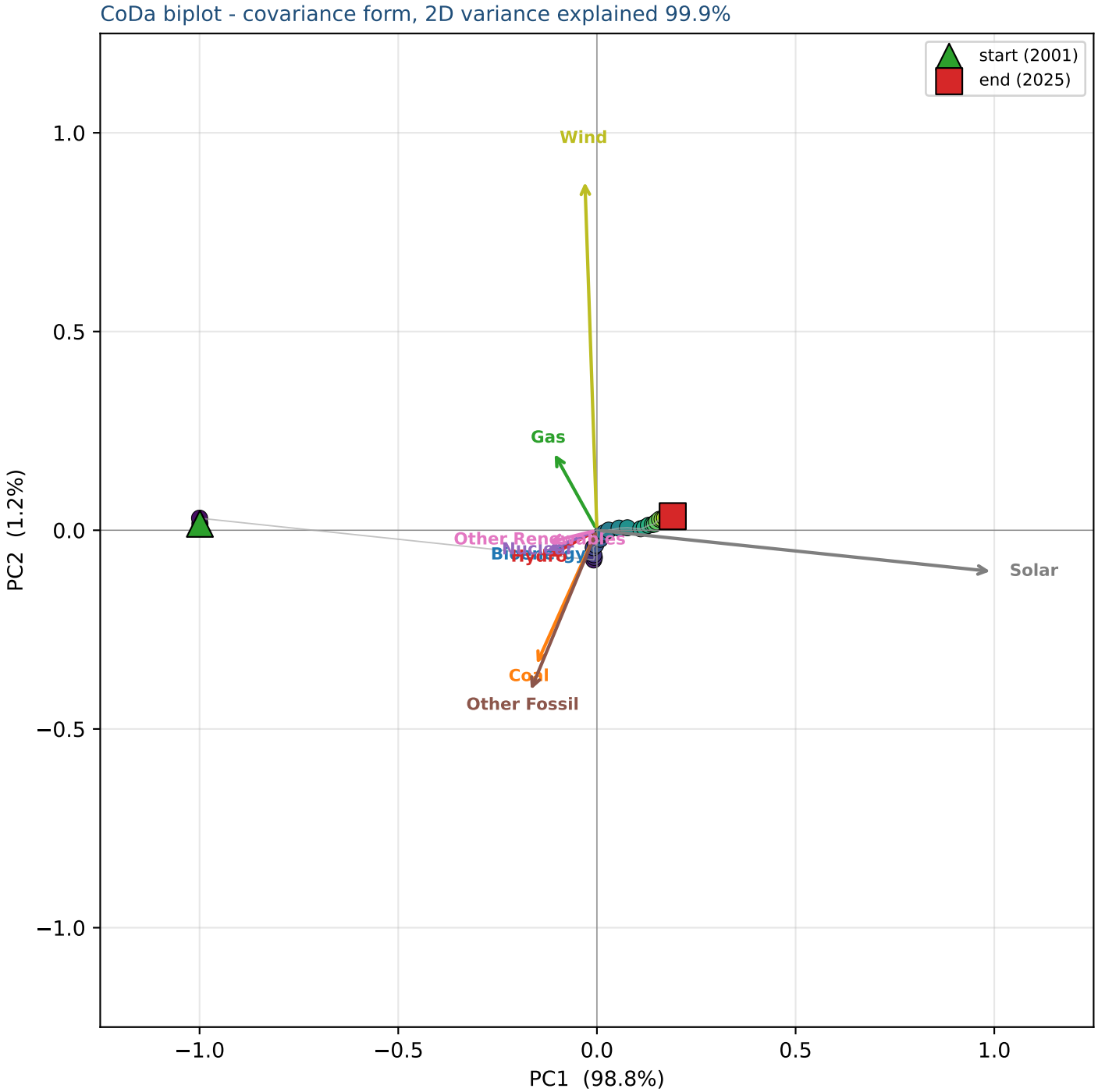
Plate 9 | CLR Biplot (CoDa biplot)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 98.8% of variance.
PC2 captures 1.2% of variance.
Total 2D plane:  99.9%
```

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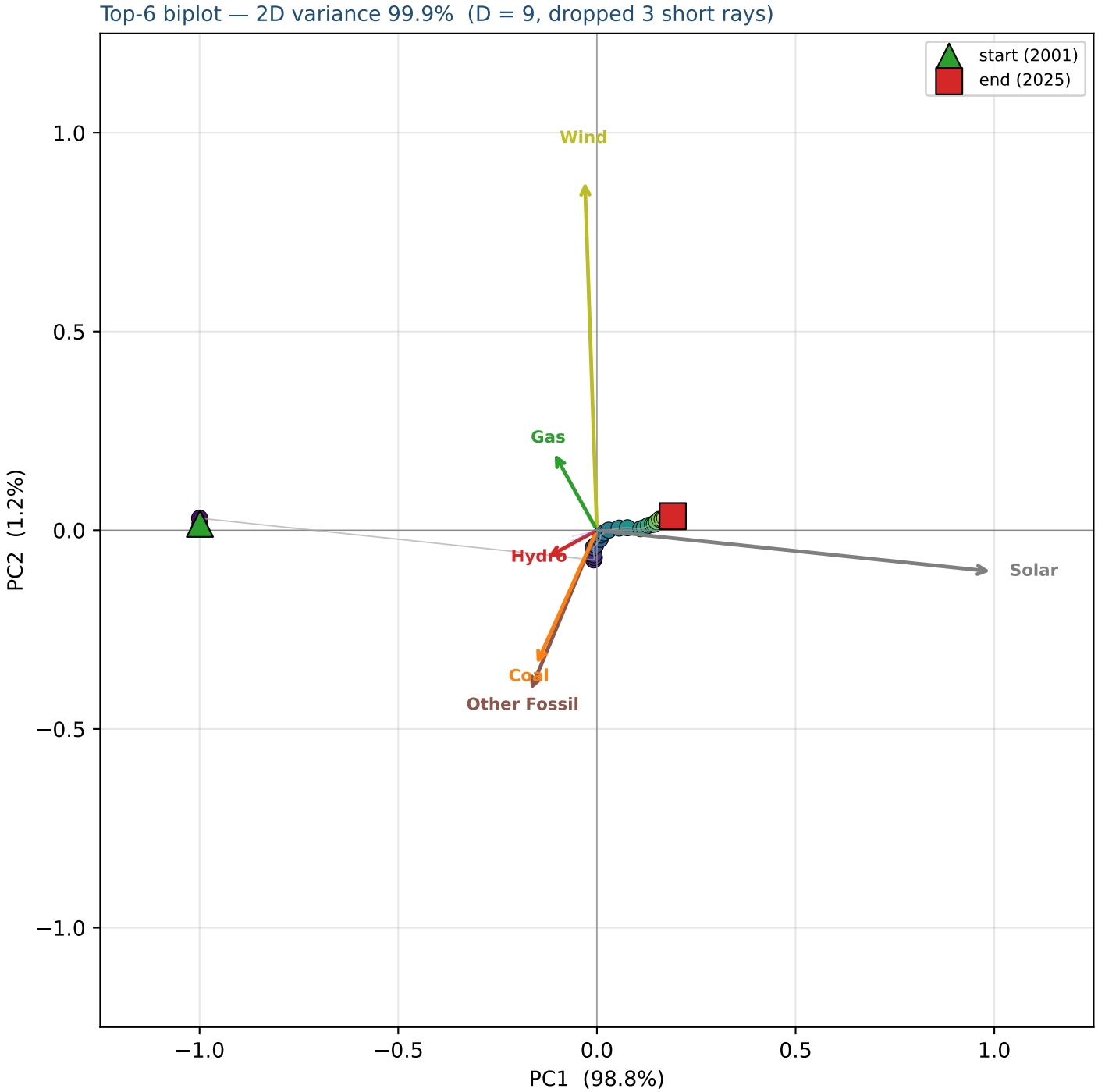
Plate 10 | CLR Biplot — top-6 carriers by loading magnitude

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Loading magnitudes $\sqrt{(G_1^2 + G_2^2)}$:

+ Solar	0.9428
+ Wind	0.8312
+ Other Fossil	0.4167
+ Coal	0.3550
+ Gas	0.2151
+ Hydro	0.1413

Dropped (faint stubs only):

Bioenergy	0.1403
Nuclear	0.1369
Other Renewables	0.1264

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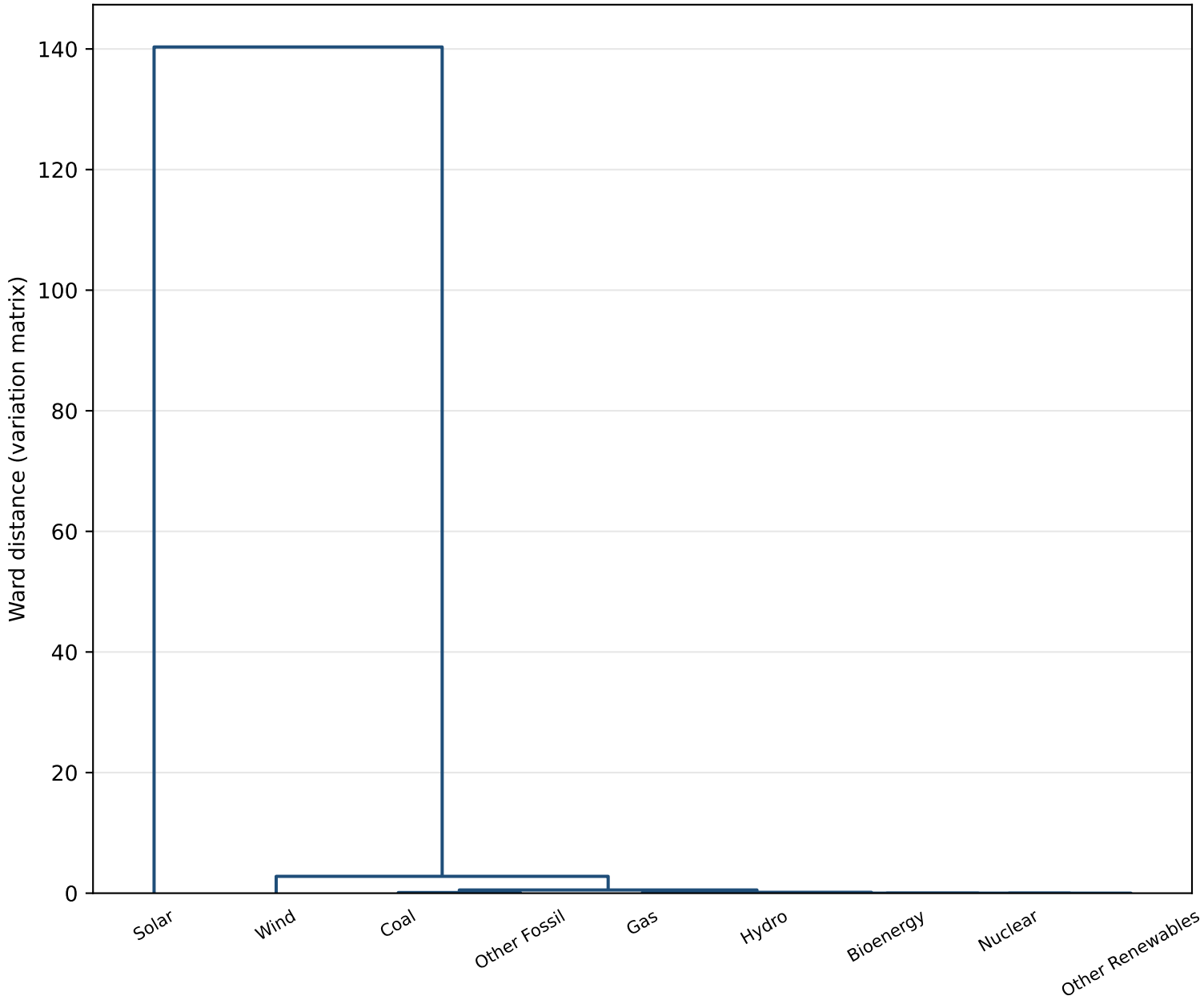
Plate 11 | Balance-Dendrogram (Ward on variation matrix)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 9 carriers → D-1 = 8 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

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Plate 12 | Sequential Binary Partition (SBP) Matrix

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, - = denominator, . = not in this balance

	b1	b2	b3	b4	b5	b6	b7	b8
energy	.	+	-	.	-	-	-	-
Coal	.	.	.	+	.	+	-	-
Gas	.	.	.	.	+	-	-	-
Hydro	.	.	+	.	-	-	-	-
uclear	+	-	-	.	-	-	-	-
Fossil	.	.	.	-	.	+	-	-
wables	-	-	-	.	-	-	-	-
Solar	.	.	.	.	.	.	.	+
Wind	.	.	.	.	.	.	+	-

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) | D = 9; D-1 = 8 balances.

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Plate 13 | Hs (Higgins scale) trajectory — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

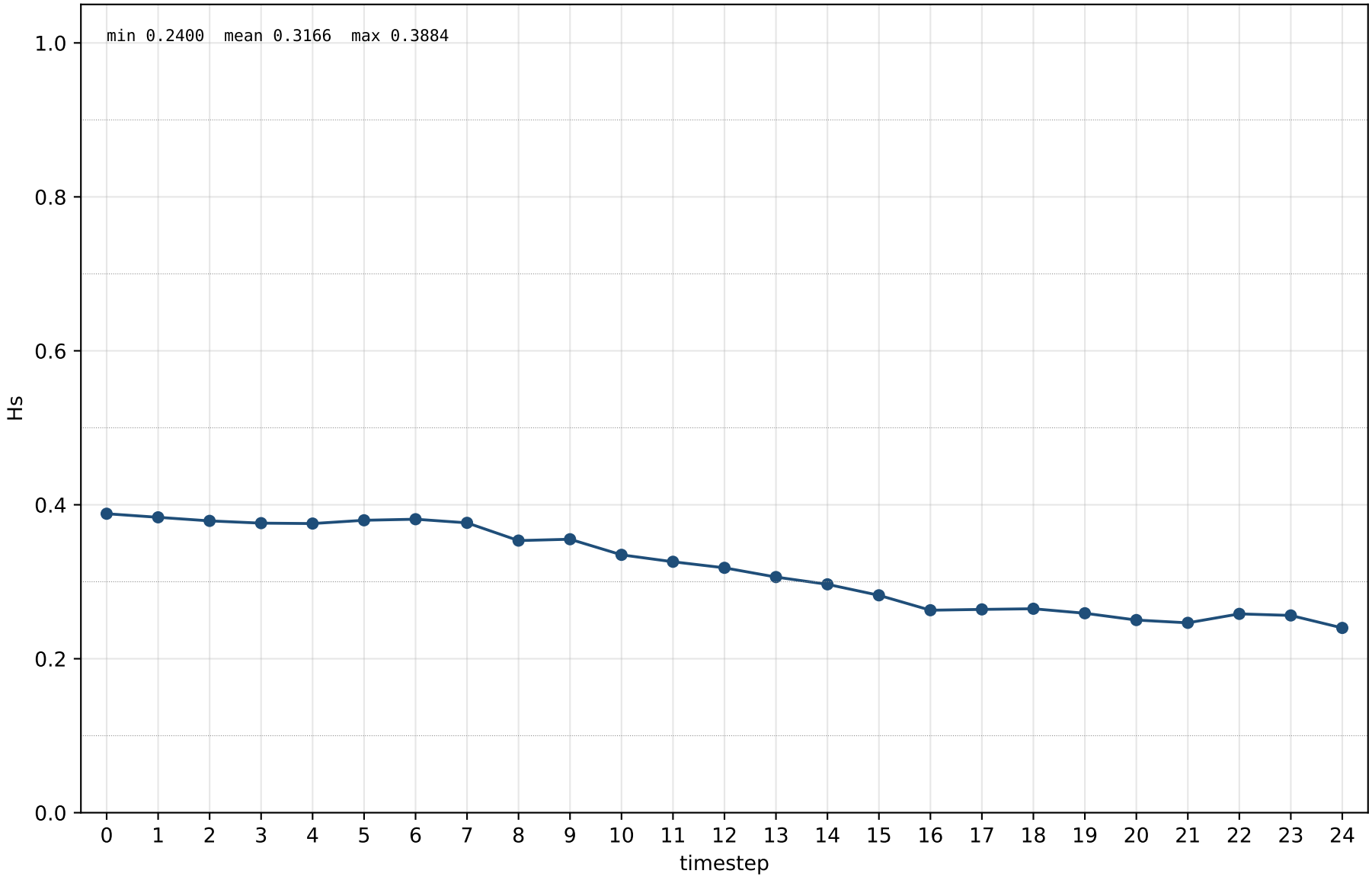

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Plate 14 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

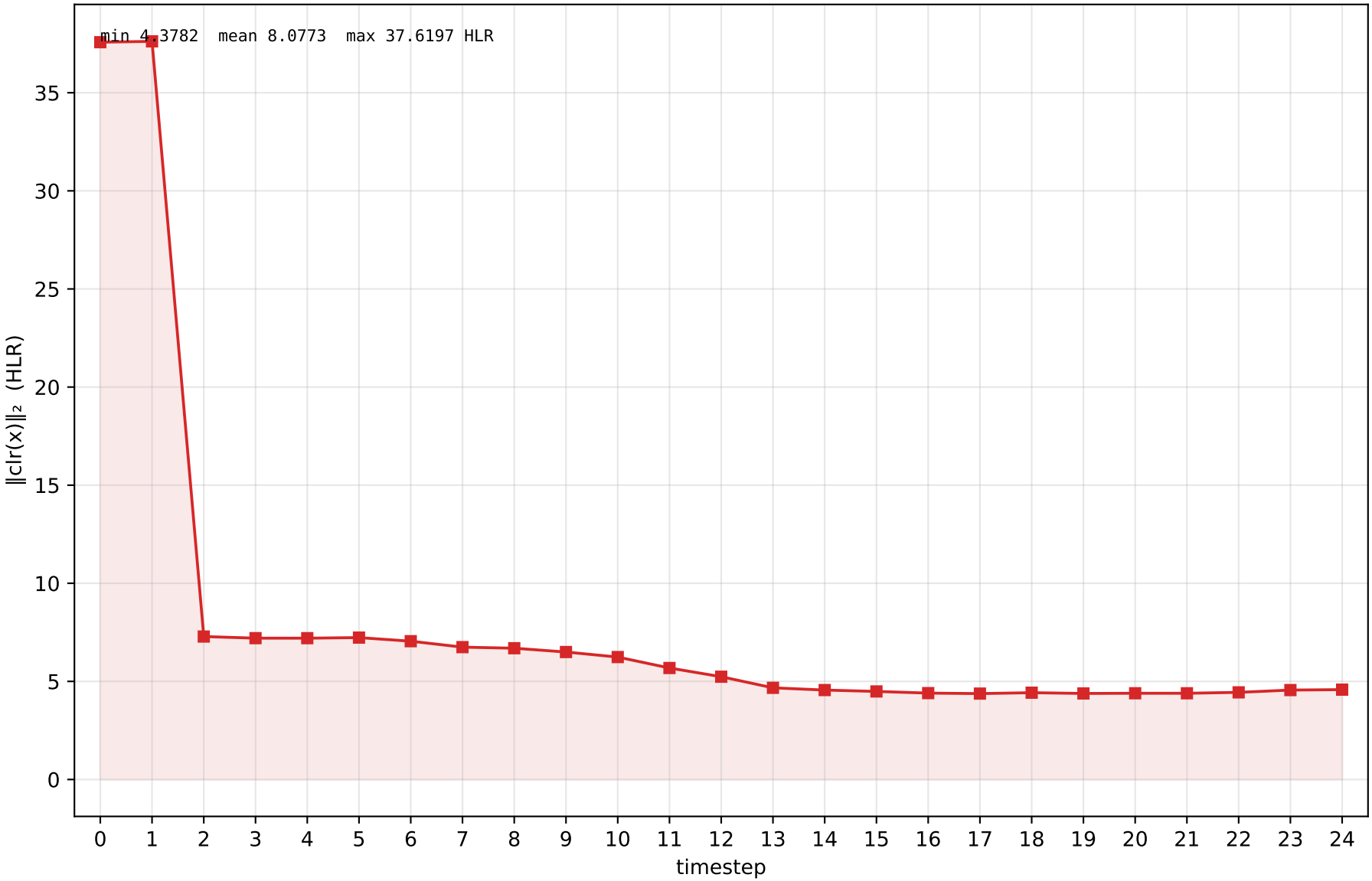

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Plate 15 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

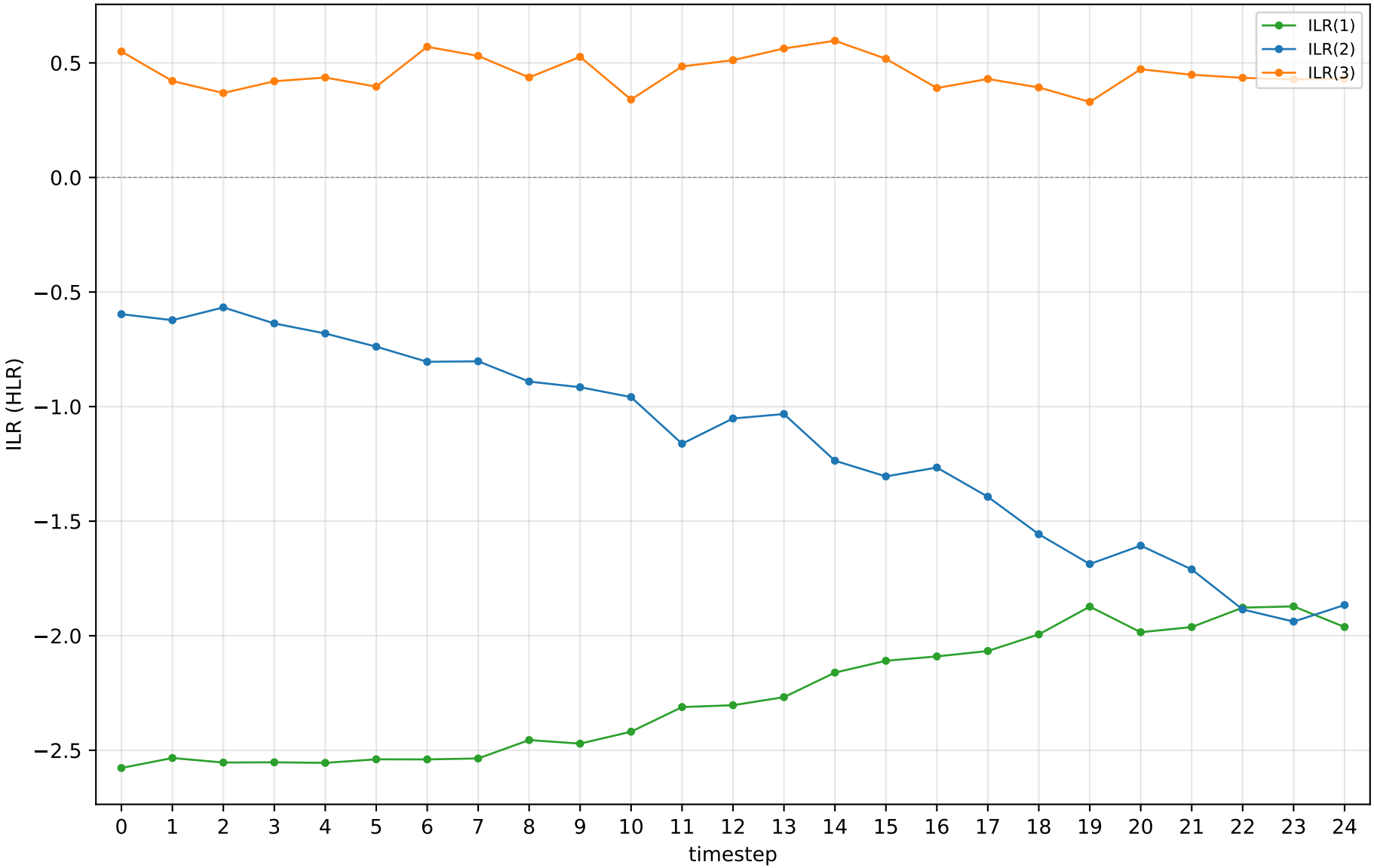

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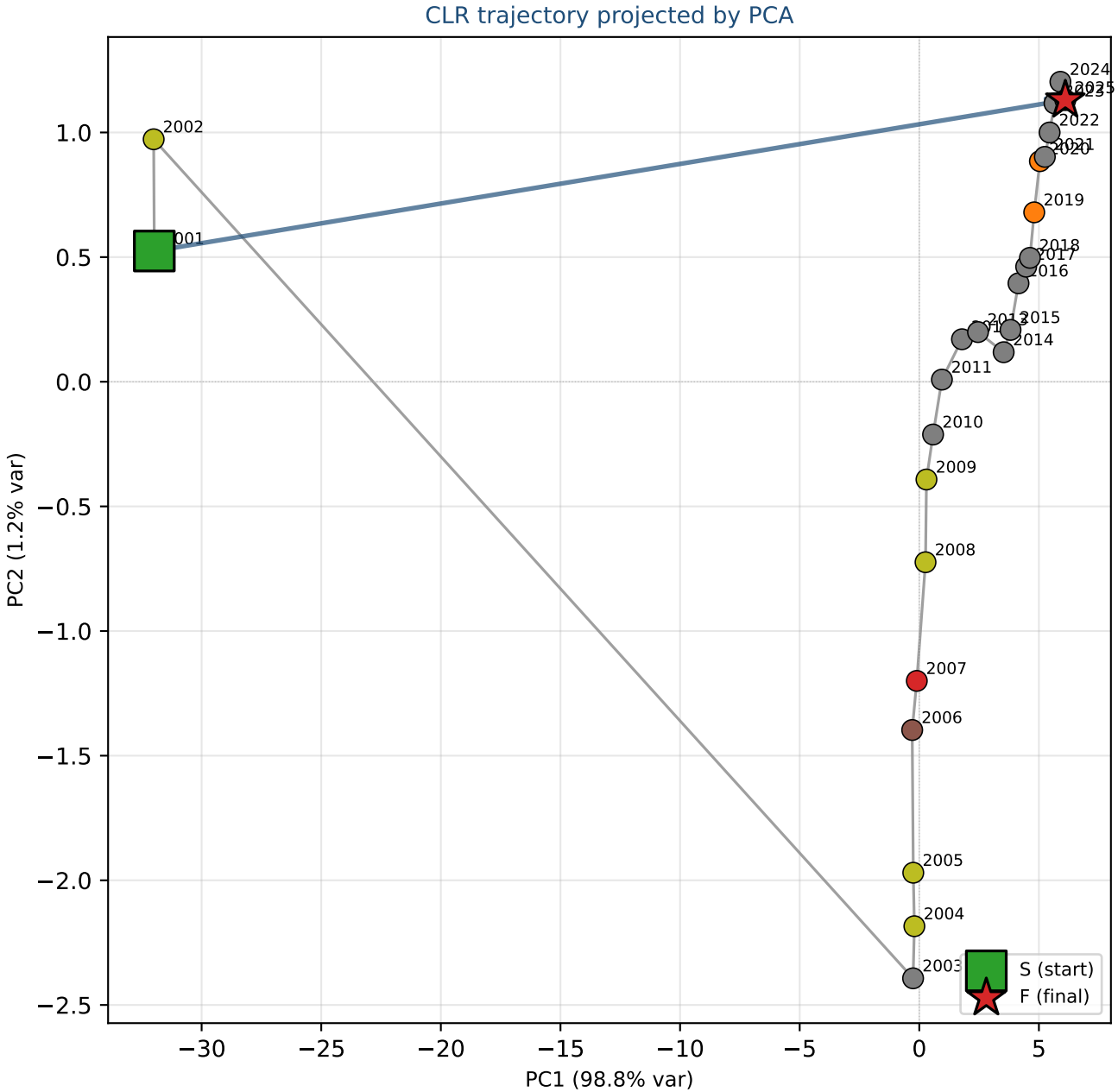
Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Navigation metrics (V_net = h_F − h_S)

```
h_start label    : 2001
h_final label    : 2025

net_distance     : 38.0885 HLR
path_length      : 41.8379 HLR
course_directness: 0.9104
  (1.0 = straight line, 0 = pure looping)

dynamic range S  : 42.0905 HLR
dynamic range F  : 4.6326 HLR

Reading guide:
  tight cluster  → stable composition
  long excursion → structural shift
  sharp turn     → regime transition
  loop           → cyclic behaviour
  spiral         → systematic drift
```

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Plate 17 | Polar bearing rose (4 sample timesteps)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

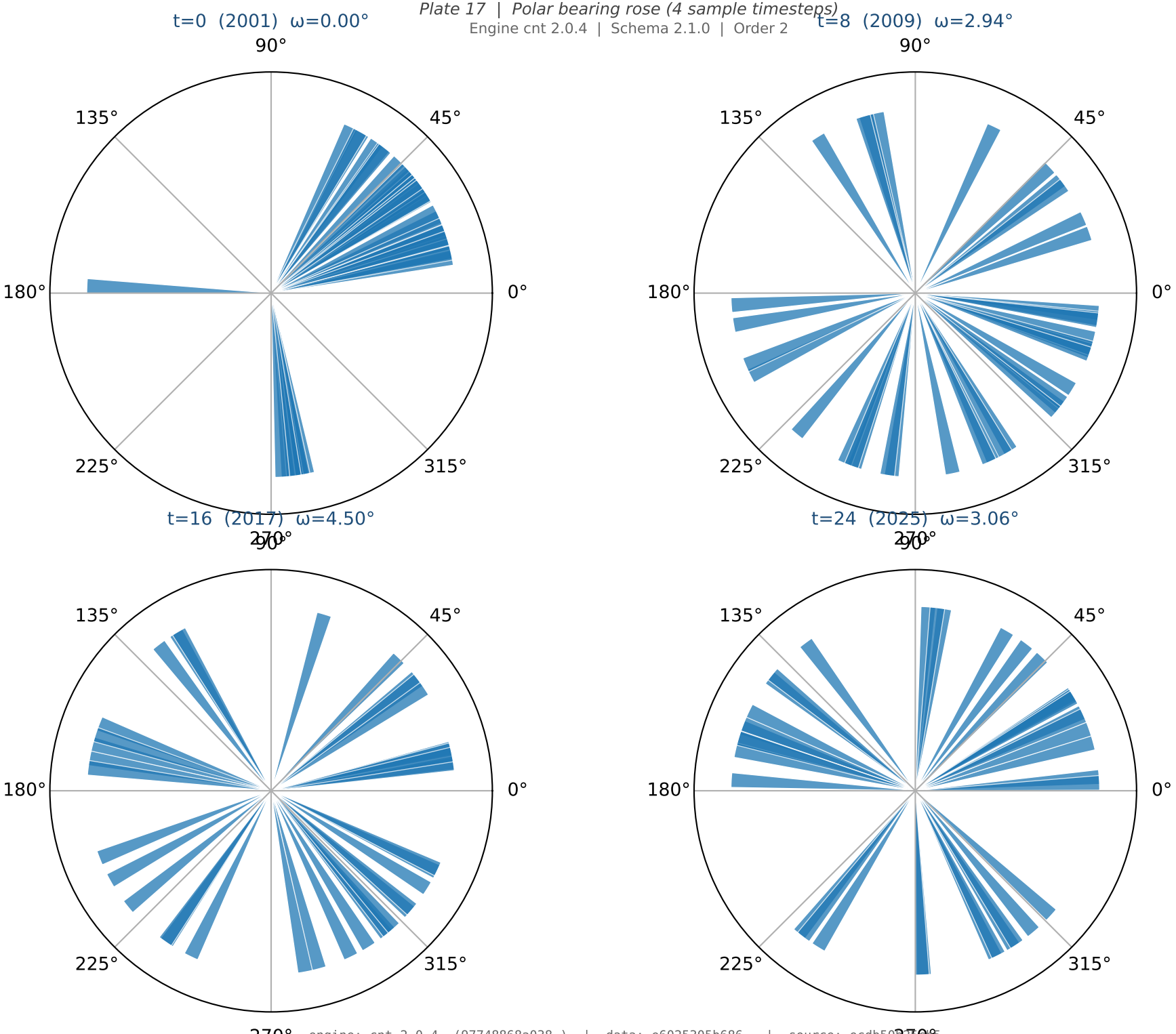

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Plate 18 | Bearing heatmap T × C(D,2) pairs

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

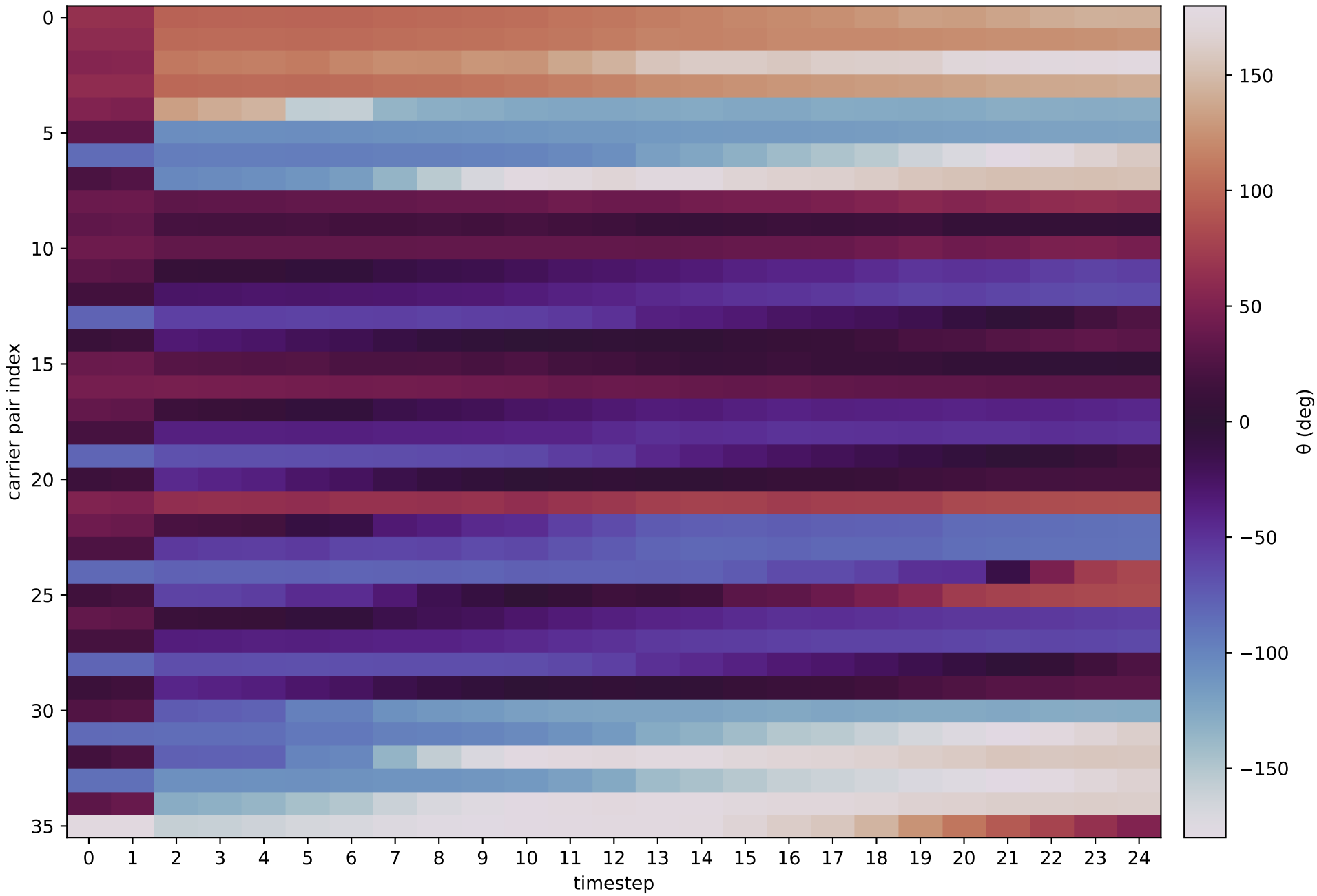

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Plate 19 | ω angular velocity trajectory

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

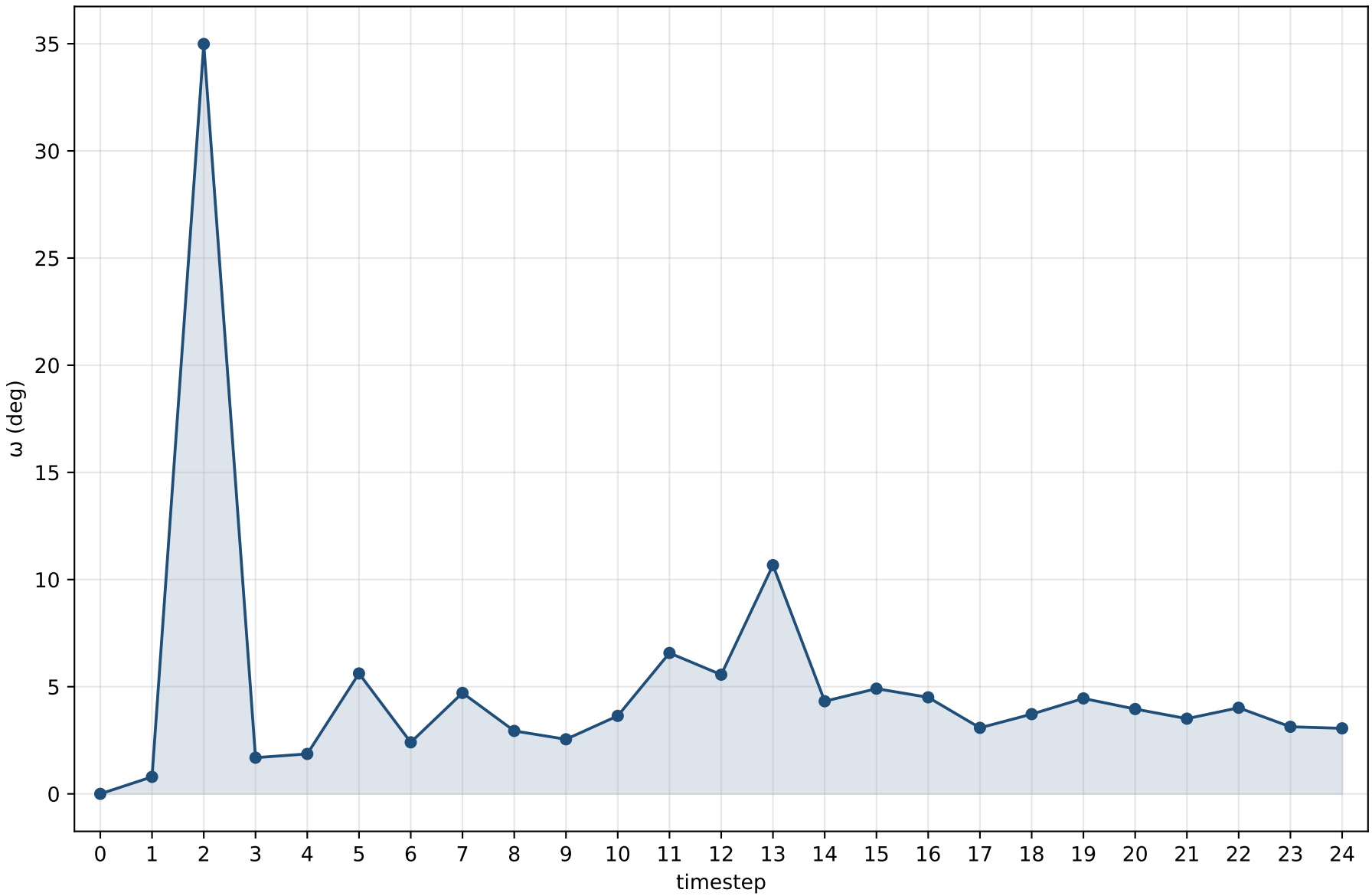

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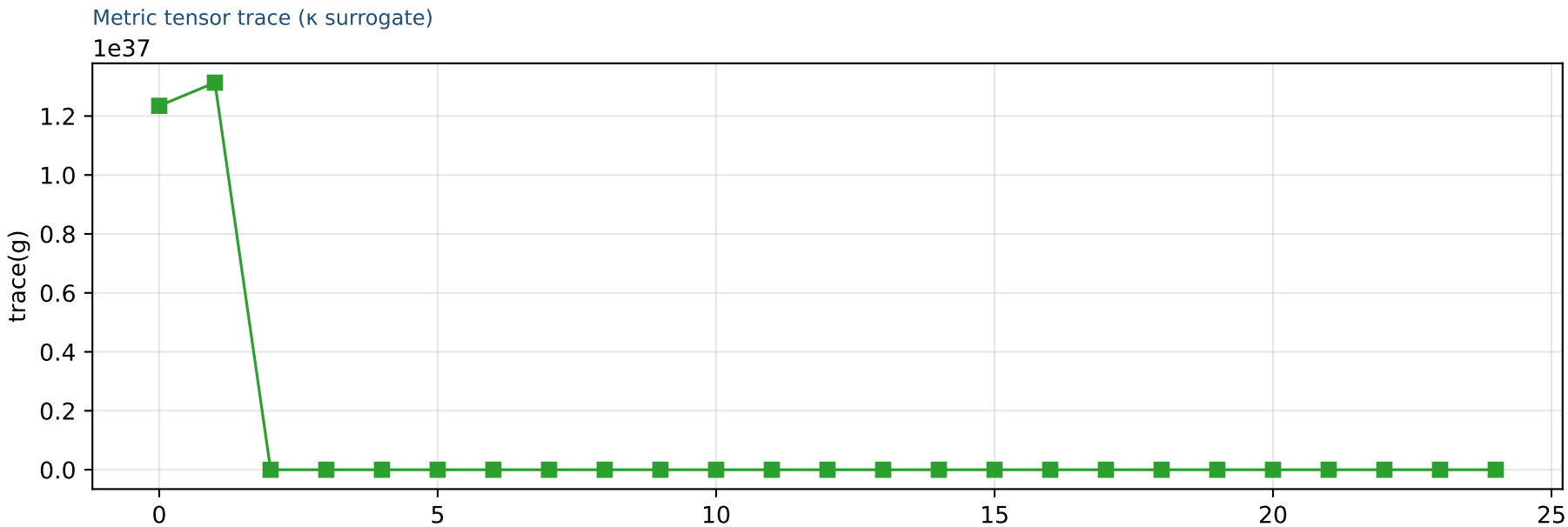
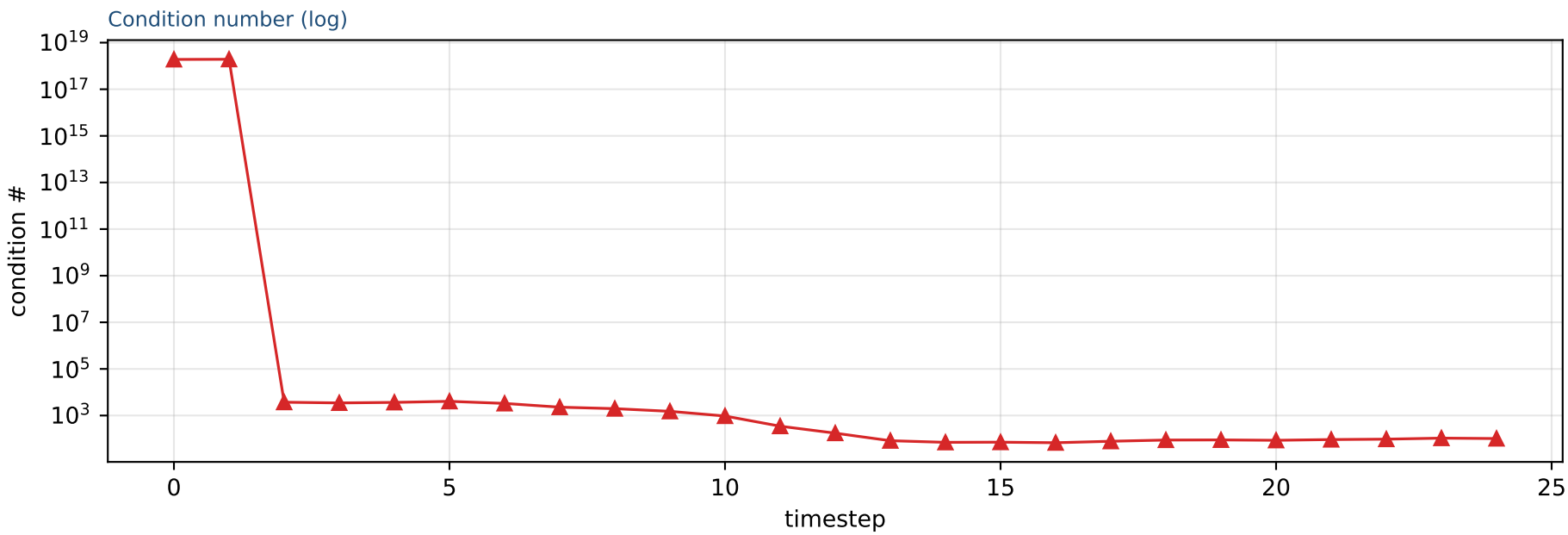
Plate 20 | Metric tensor trace + condition number

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2





Compositional system geometry and dynamics for, United States — EMBER electricity generation 2001-2025

Plate 21 | Helmsman σ lineage timeline

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

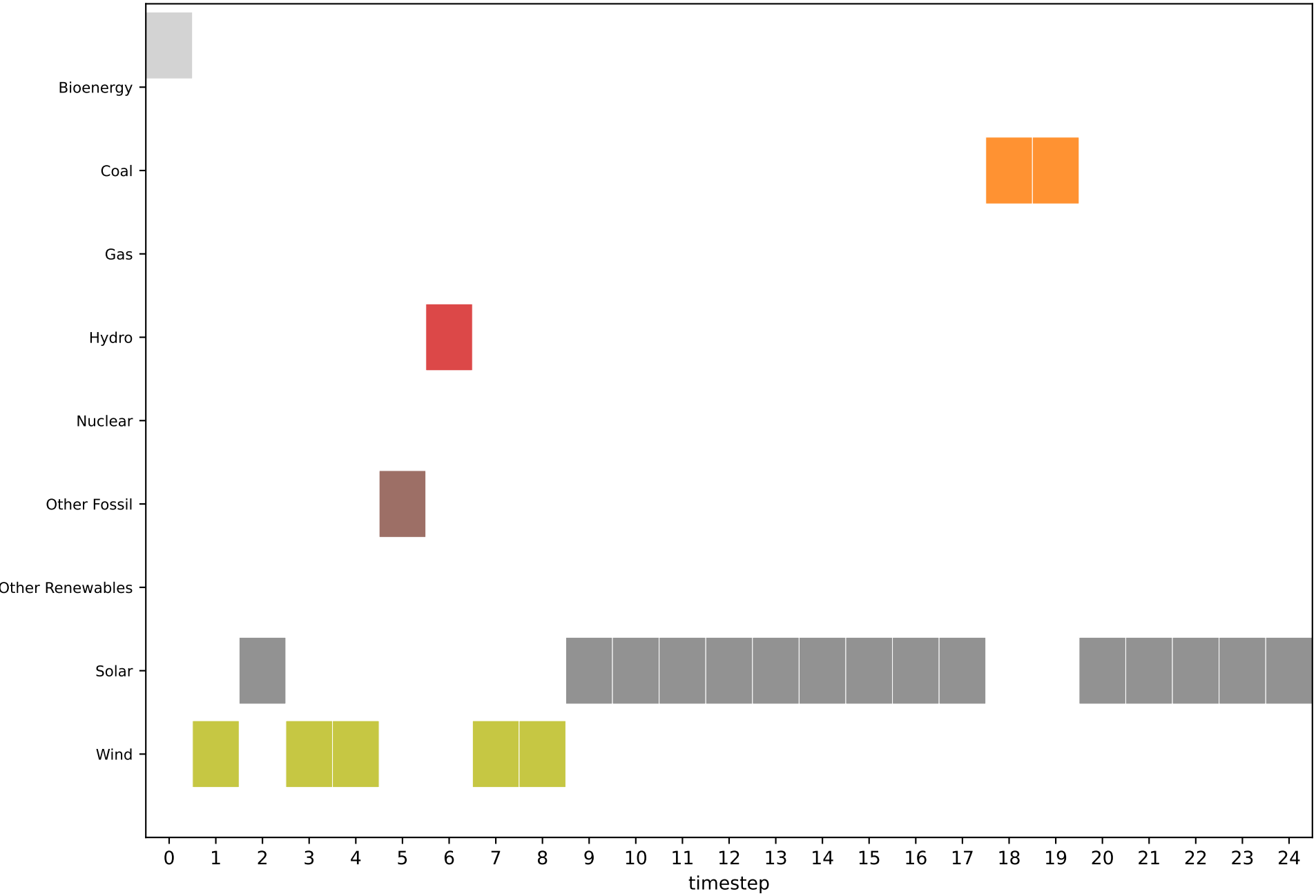


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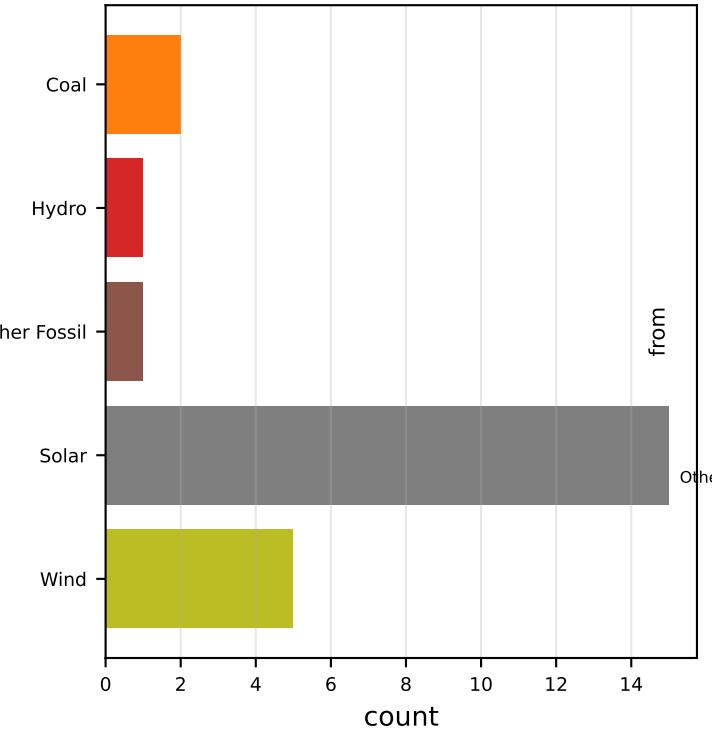
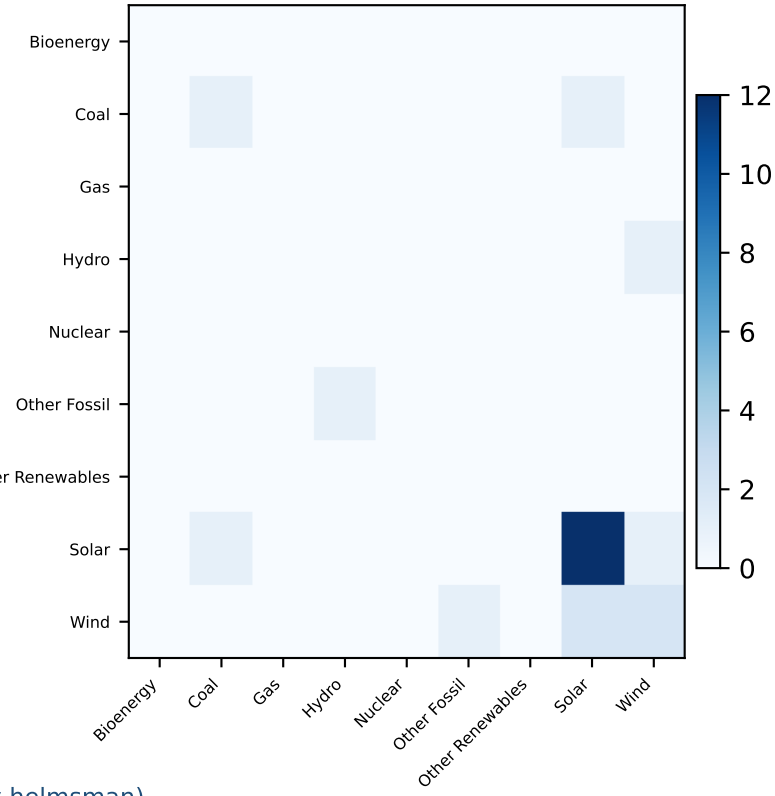
Plate 22 | Helmsman σ — frequency · transitions · per-year displacement

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Helmsman frequency



Transition matrix



Helmsman displacement per year (bar coloured by helmsman)

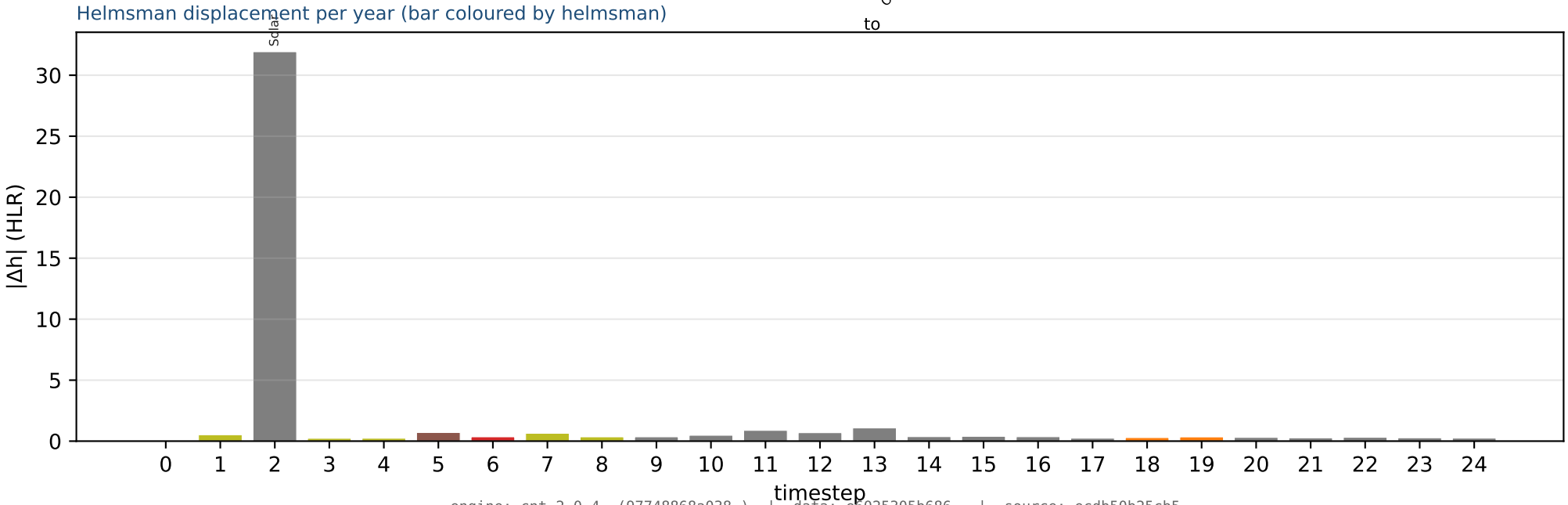


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Plate 23 | Ring class evolution + Hs

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

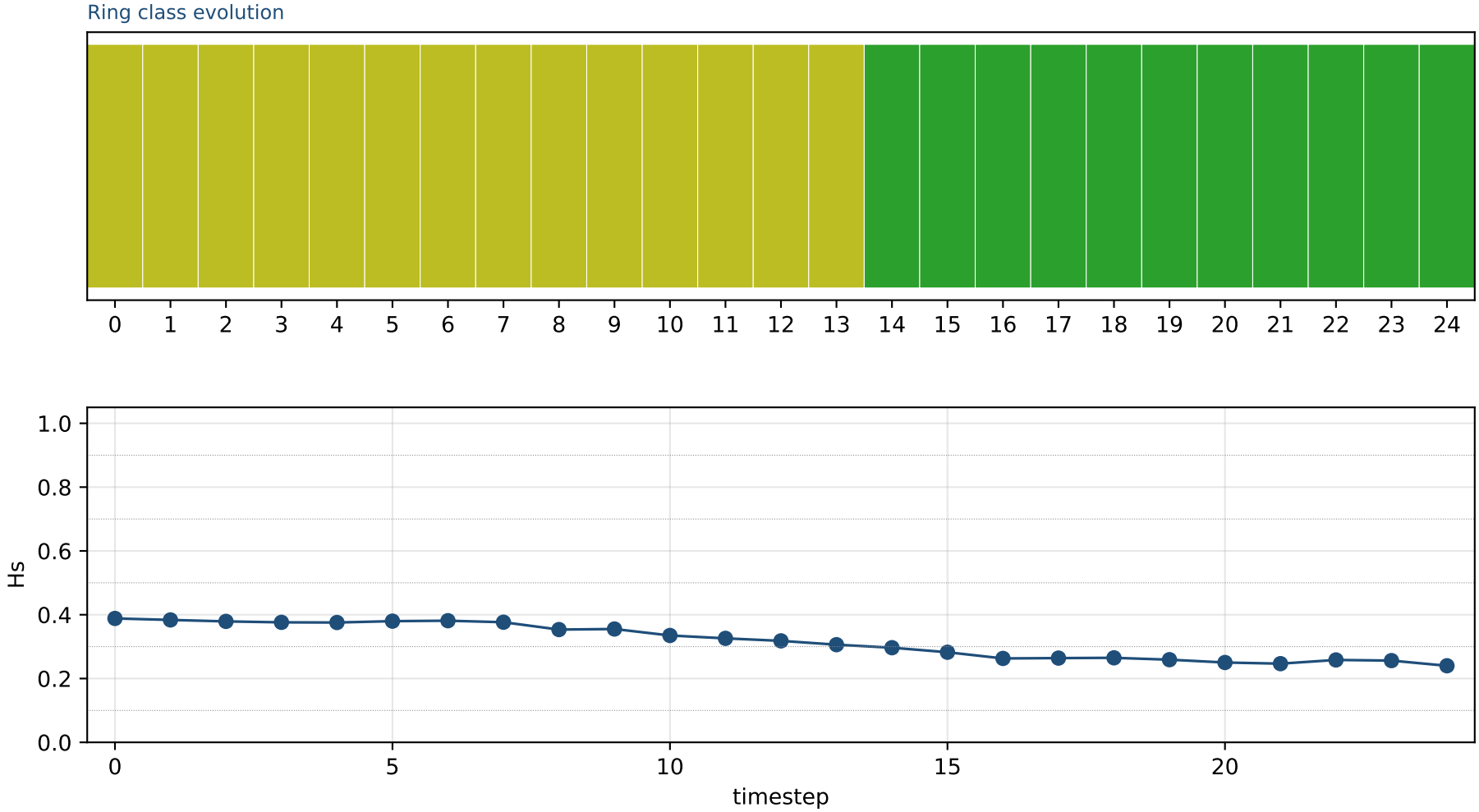

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Plate 24 | Lock event timeline (carriers × timesteps)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

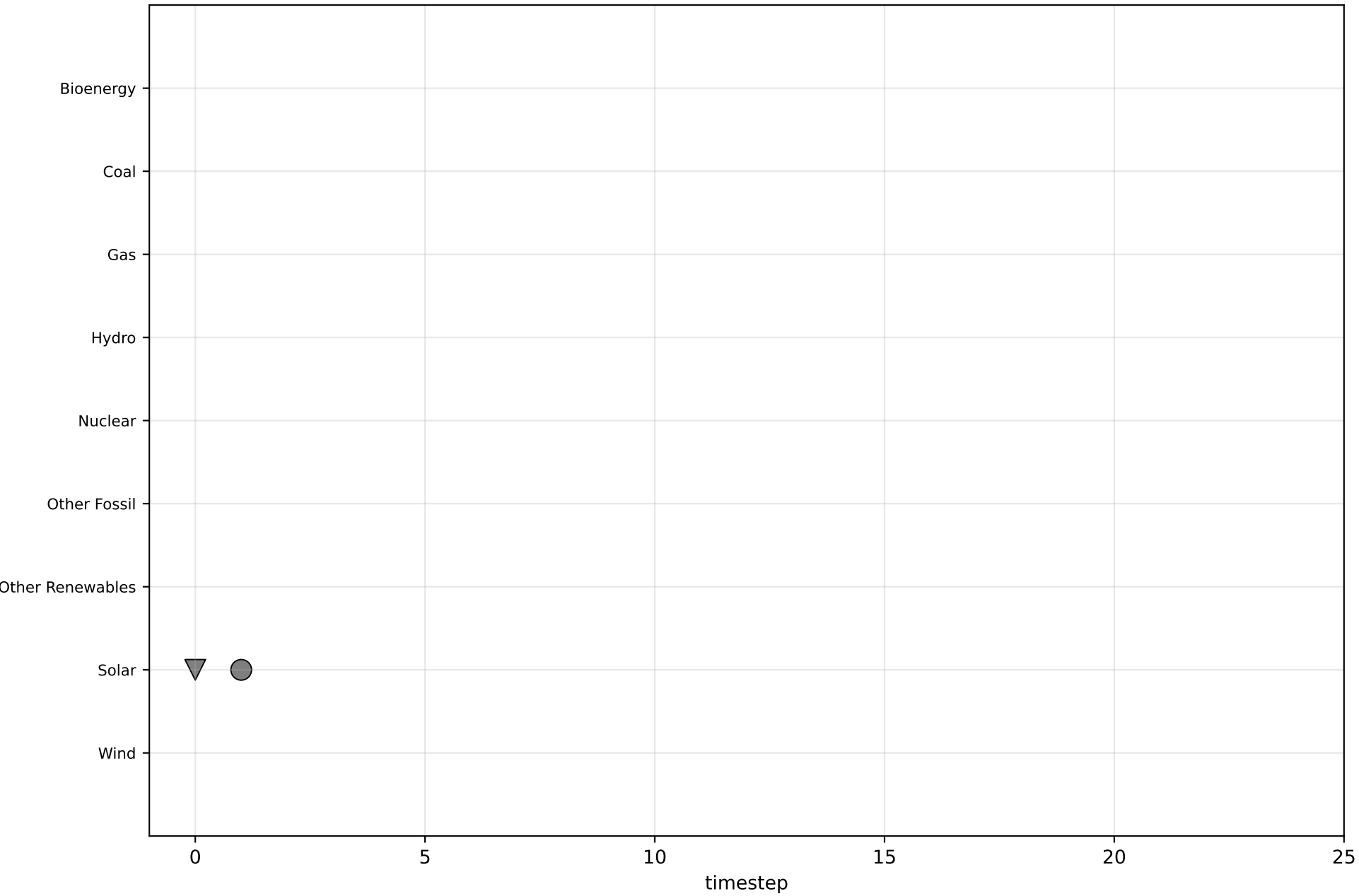

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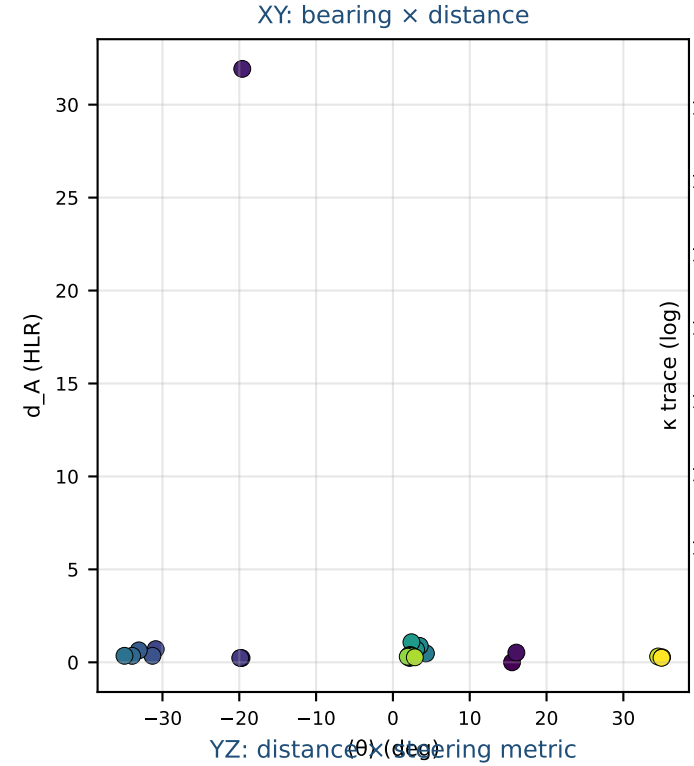
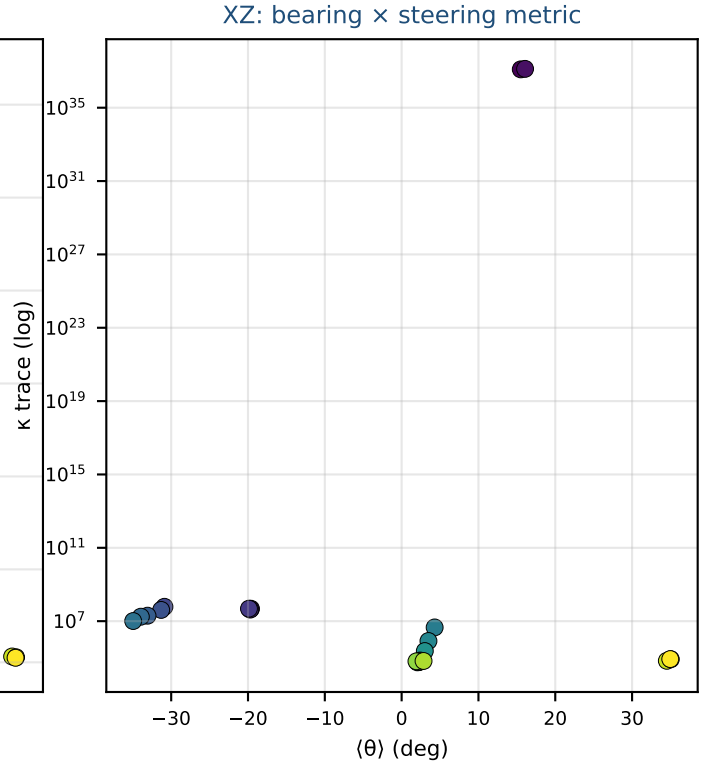
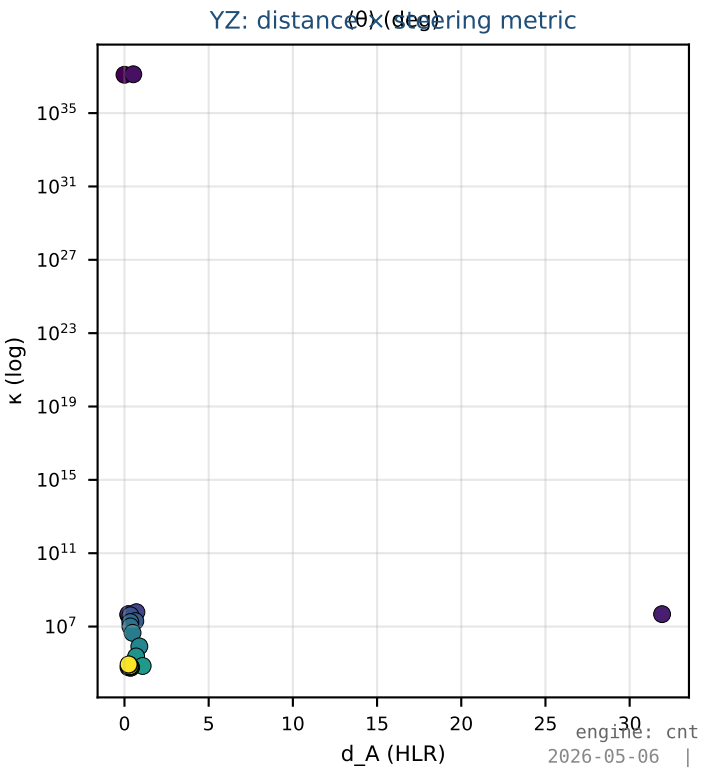
Plate 25 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2







CBS info panel (Math Handbook Ch 16)

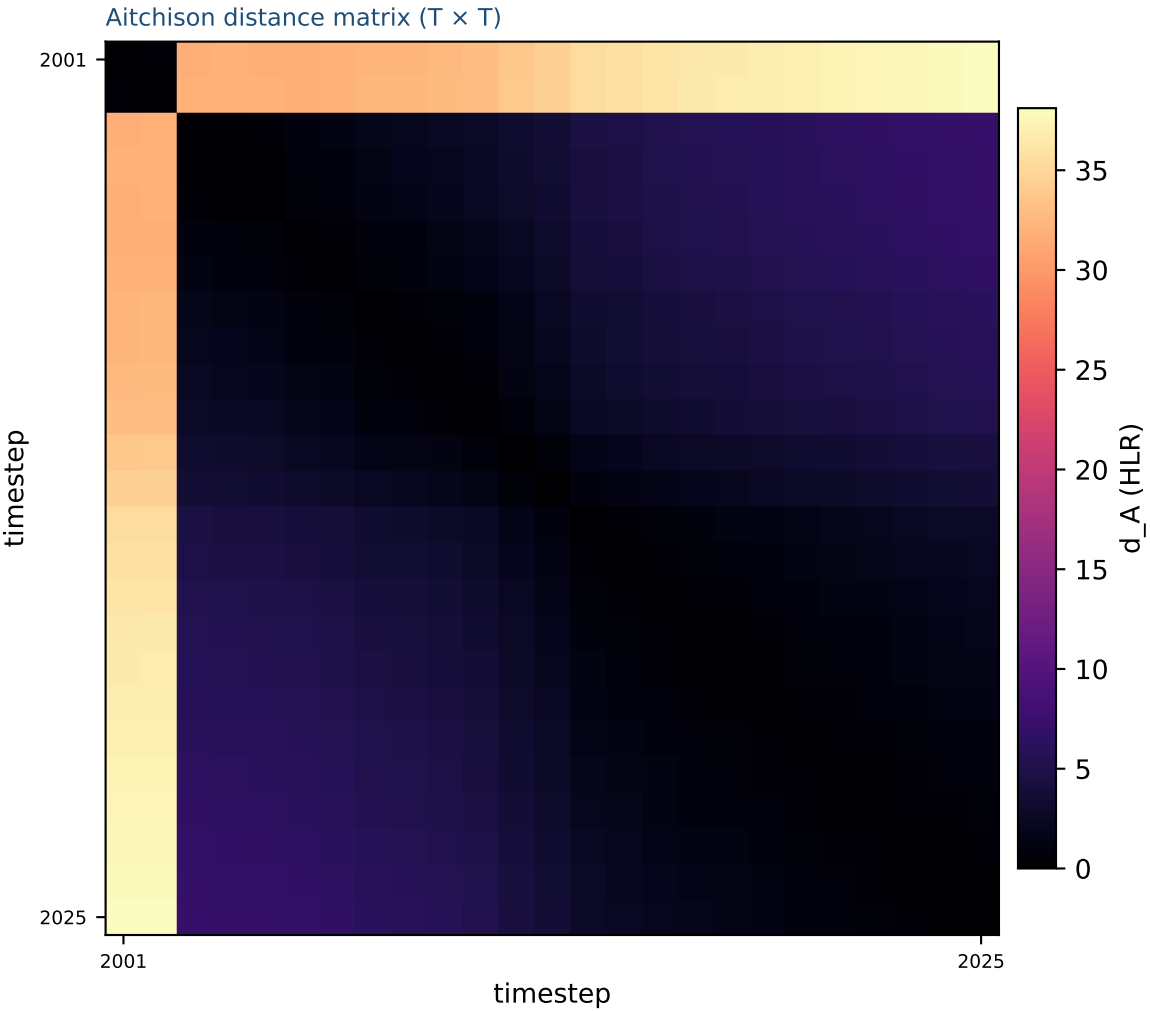
```
⟨θ⟩ range  : [-34.9, 35.0]°
d_A range  : [0.0000, 31.9255] HLR
κ range    : [5.69e+04, 1.31e+37]
Hs range   : [0.2400, 0.3884]
ω max      : 34.99°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

Compositional system geometry and dynamics for, United States — EMBER electricity generation 2001-2025

Plate 26 | Pairwise divergence — distance matrix + top-15 ranking

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

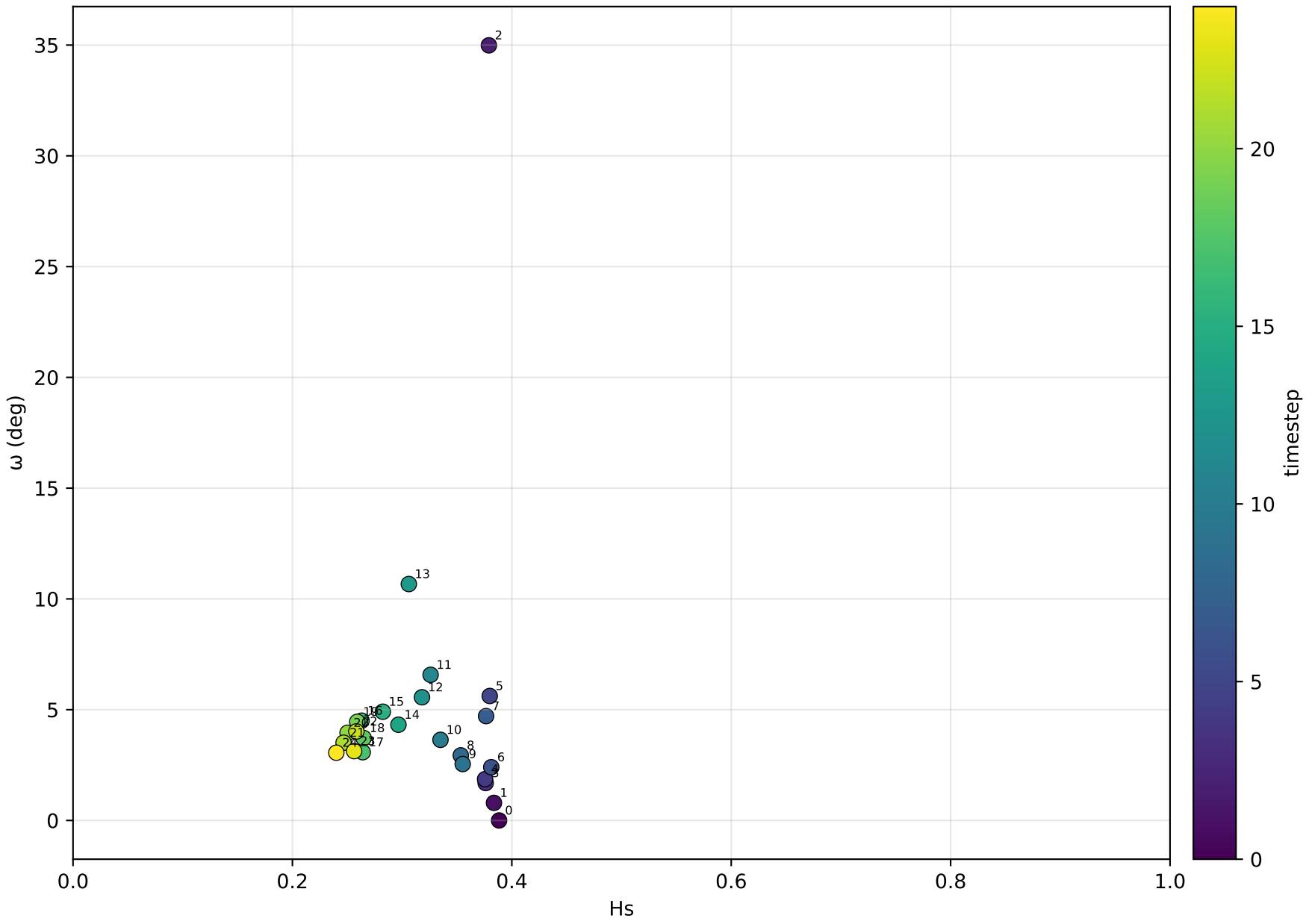


rank	A	B	d_A
1	2002	2025	38.1117
2	2001	2025	38.0885
3	2002	2024	37.9059
4	2001	2024	37.8834
5	2002	2023	37.6593
6	2001	2023	37.6357
7	2002	2022	37.4542
8	2001	2022	37.4295
9	2002	2021	37.2539
10	2001	2021	37.2288
11	2002	2020	37.0480
12	2001	2020	37.0230
13	2002	2019	36.8108
14	2001	2019	36.7835
15	2002	2018	36.6338

Compositional system geometry and dynamics for, United States — EMBER electricity generation 2001-2025

Plate 27 | ω vs Hs scatter (XY test)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



```
T = 25,  D = 9
carriers: Bioenergy, Coal, Gas, Hydro, Nuclear, Other Fossil, Other Renewables, Solar, Wind

Navigation (System Course Plot):
  net_distance     = 38.0885 HLR
  path_length      = 41.8379 HLR
  course_directness= 0.9104

ω (deg)        : min 0.00  mean 5.07  max 34.99
Hs             : min 0.2400  mean 0.3166  max 0.3884
Aitchison norm : min 4.3782  mean 8.0773  max 37.6197
κ trace        : min 56895.20  max 1313233417102222231215113640575028428 8.00
condition #    : min 6.80e+01  max 1.93e+18

Bearings: range [-179.21, 179.62]°,  std 89.58°
Lock events    : 2
Pairs          : 36
|r| > 0.7 pairs: 28

Helmsman frequency (top 5):
  Solar           15  (60.0%)
  Wind             5  (20.0%)
  Coal             2  (8.0%)
  (none)           1  (4.0%)
  Other Fossil     1  (4.0%)

Variation matrix trace : 0.0000
```